CODENEX · THE OTHER SIDE: HACK THE CONSTRAINT 2026 · TRAVEL & MOBILITY

RAHA

Fair, capacity-exact, offline-first recovery for stranded transit networks

Problem Statement 2: The Stranded Network

Surprise constraints: (1) no internet or third-party APIs, (3) API response structure changes

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Venue: SRMIST Chennai (TP2 712) · Platform: Manch · Internship partner: SpazorLabs

Live simulation: (add the live URL after deploying, see DEPLOY.md)

Code: https://github.com/swetank18/ANTIMATTER-RAHA

Submitted 2026-11-06 · every number in this document is reproducible: node scripts/run_experiments.mjs 200

1. Summary

When a rail line fails, thousands of riders are stranded at once. Everyone opens a cab app and the fare surges. Forty people going the same way book forty cabs. Replacement buses arrive late and in the wrong places, and the crowd crushes at the doors. **RAHA** turns that free-for-all into a coordinated recovery:

- bridging buses whose passes match seats exactly
- the most vulnerable and urgent riders served first, with a written explanation for every decision
- **Koottu** shared cabs in trusted circles, so sharing costs less than riding alone even at surge
- a geofence that brings in two-wheeler partners and essentials drops

It keeps working with the internet cut (Constraint 1) and when a data feed changes shape (Constraint 3).

We built a simulator that replays **five real disruptions** (Chennai, New Delhi, Mumbai, Singapore and Madrid) on islands made from real OpenStreetMap data. It measures RAHA against four honest alternatives over 200 seeded runs each.

Headline (Chennai, Tambaram station, 28 Feb 2026, 200 runs): the worst-off rider waited 224 min under free-for-all, 213 min under a first-come-first-served app and **199 min under RAHA**. Failed boarding attempts at bus doors fell from 10,354 to **0**. Deadlines met rose from 85% to **99%**. Koottu carried **3.41 riders per cab** at 134 rupees each versus 406 alone, and nobody paid more than their solo fare.

2. The original problem (PS2: The Stranded Network)

Transit disruptions strand large crowds in minutes, while recovery capacity (buses, cabs, information) arrives in hours and is allocated by whoever pushes, pays or taps fastest. We studied five recent, documented incidents:

Incident (local date)	What happened (sourced)	Why it happened
Chennai Tambaram station 2026-02-28	A signal fails south of Tambaram. Trains stop on one track. The promised buses never come.	Footfall about 1.5 lakh a day with lagging infrastructure; services already cut from 204 to 164; few agencies keep bus-bridging reserves (Pender et al., 2013).
New Delhi New Delhi Railway Station 2025-02-15	Maha Kumbh rush meets delayed trains and confusing announcements. 18 people die on FOB-3 stairs.	Tickets not linked to seats; similar train names caused confusion; a single staircase between platforms (Fruin's FIST model: Force, Information, Space, Time).
Mumbai Kurla station 2025-08-19	Heavy rain floods tracks. Central Railway suspends CSMT-Thane. Thousands stranded across stations.	Drains handle about 25 mm/h at low tide; only 3 of 45 outfalls have floodgates; footbridges too narrow (Elphinstone 2017: 2.4 m, 23 dead).
Singapore Jurong East station 2024-09-25	A faulty train damages track and power on the East-West Line. No trains Jurong East to Buona Vista for six days.	Axle-box failure after an overhaul interval was extended without assessment; bridging point an 8-minute walk away; riders confused between bus services.
Madrid Atocha station 2025-04-28	At 12:33 the Iberian grid collapses. Trains stop, the metro is evacuated, phones barely work.	Grid voltage-control failure (ENTSO-E); every cloud-dependent app and payment system stopped together; mobile networks degraded.

3. Our solution: RAHA

- **Bridging with capacity-exact passes.** Riders are grouped by real destination corridor. Buses go where the most urgent riders wait, and accessible buses go where wheelchair users wait. A pass exists only if a seat exists. Pass holders are released from a holding area in waves at no more than 1 person per square metre (Fruin level D).
- **Fair, explainable priority.** Score = 35·V (vulnerability) + 30·D (deadline urgency) + 25·W (wait, which never decreases) – 10·C (detour). Willingness to pay is never an input. A refused rider receives a refusal proof: score, cutoff and rank, sealed with a SHA-256 record hash.
- **Koottu shared cabs.** Surge is charged per car, not per person, so sharing at the 2× legal cap costs less than riding alone at the normal fare. Groups form inside trusted circles (the Zimride model that became Lyft). The cheapest of the group's phone quotes books the cab. The split uses the Shapley value (airport game), and the rule

that nobody pays more than their solo fare is enforced directly in code. Fares are checked against the 2× cap of the Motor Vehicle Aggregator Guidelines 2025.

- **Geofence, two-wheelers and essentials.** Riders, cabs and delivery partners inside the geofence are alerted. Bike-taxi or delivery partners take urgent solo riders where state law allows (Karnataka banned bike taxis in 2025). Riders waiting more than 45 minutes are flagged for water and food drops.

4. The surprise constraints and how we adapted

Constraint 1: No internet, no third-party APIs

Card: All external internet connectivity and third-party API access become unavailable during the challenge. The application must continue its essential user journey using information and components available within the local environment.

How RAHA adapts (product)

- The station edge node (on UPS power) runs grouping, scoring and seat allocation with no cloud dependency (local-first).
- Passes are signed on the edge node and verified on driver handhelds with no network.
- Riders without signal check in at kiosks or with staff tokens; Koottu still books a cab if any one phone in the group has signal.
- Timetables, fleet registry and maps are cached locally; no maps API, no ride-hail API, no LLM.

How the simulator implements and proves it

- Madrid blackout scenario: the real 28 April 2025 blackout at 12:33. Left: a cloud-only coordinator collapses into free-for-all. Right: RAHA on the edge node keeps allocating.
- Every 200-seed experiment sweeps phone reach 100%, 70%, 30% and 0%, comparing cloud and local architectures (Chennai and Madrid).
- The simulator obeys the card too: three.js, fonts, maps and data are bundled; the EXTERNAL REQUESTS badge counts and blocks any outside call and stays at 0.

Constraint 3: An API changes its response structure

Card: A previously working API suddenly changes its response structure. A field may be renamed, moved into a nested object, change type, or become optional.

How RAHA adapts (product)

- One adapter sits between every feed and the allocator: alias paths for renamed and nested fields, coercion for type changes, last-known-good and a local vehicle registry for missing fields.
- Invariant checks catch semantic drift (same field, new meaning), such as an ETA that silently becomes epoch milliseconds or a capacity that becomes crush load.
- Fail safe: if a seat count cannot be trusted, no passes are issued for that vehicle, so nobody is overbooked.

How the simulator implements and proves it

- Singapore line-closure scenario: the bus feed mutates at 10:30 mid-incident; pick any of six mutations (rename, nest, retype, optional, garbage, semantic) and watch both sides re-run live.
- The JSON panel shows the real message, what a naive parser reads (NaN, a throw, a wrong count) and what RAHA's adapter reads.
- Every mutation class is measured over 200 seeds, naive versus adapter (Chennai and Singapore): empty departures, turned away, worst wait.

5. How we tested: the RAHA simulator

- **Real geography and timelines.** OpenStreetMap geometry for each hub (buildings, roads, rail, platforms, footbridges, bus stops). Destination stations at their real coordinates and distances. Timetables from the Indian Railways and Chennai MTC GTFS feeds. Incident timelines from cited reports. Demographics from Census 2011.
- **Data tiers.** Every input is T1 (observed), T2 (derived by formula), T3 (research constant) or T4 (assumption). T4 values are stated and stress-tested. Individual riders are sampled from real counts and distributions, because no per-person data exists.
- **One deterministic engine.** A discrete-event simulation (src/sim/engine.js) drives both the 3D replay and the statistics. Paired comparisons use the same riders and fleet per seed.
- **Five policies.** Free-for-all (status quo), digital first-come-first-served, market pricing (highest bidder), pure throughput (ignores need), and RAHA.
- **Checks.** A physical sanity bound (clearance ≥ (waves – 1) × shortest round trip); sensitivity runs with door-crush delay at 0, demand ×0.5 and ×2, and fleet ×0.5; a Constraint 1 phone-reach sweep; and six Constraint 3 mutation classes.

6. Results of testing (mean over 200 seeds, 95% bootstrap interval)

Chennai (Tambaram station) · 2026-02-28 · N = 1,890 riders · sanity bound 61 min

Metric	Free-for-all	FCFS app	Market	Throughput	RAHA
Worst-off wait (min)	224 [222–226]	213 [210–217]	219 [215–223]	217 [214–220]	199 [194–204]
Mean wait (min)	19.5 [19.4–19.6]	14.6 [14.5–14.6]	14.9 [14.8–14.9]	14.6 [14.5–14.6]	15.2 [15.1–15.4]
Wheelchair p95 wait (min)	220 [218–222]	209 [205–212]	214 [209–218]	213 [209–216]	192 [187–197]
Deadlines met (share)	0.85 [0.85–0.85]	0.92 [0.92–0.92]	0.91 [0.91–0.91]	0.92 [0.92–0.92]	0.99 [0.99–0.99]
Turned away at doors	10,354 [10,267–10,447]	0 [0–0]	0 [0–0]	0 [0–0]	0 [0–0]
Peak door density (p/m²)	4.24 [4.18–4.30]	1.75 [1.75–1.75]	1.75 [1.75–1.75]	1.75 [1.75–1.75]	1.00 [1.00–1.00]
Unserved at horizon	5 [4–5]	5 [4–5]	4 [4–5]	5 [5–5]	3 [2–3]
Riders per cab	1.00 [1.00–1.00]	1.00 [1.00–1.00]	1.00 [1.00–1.00]	1.00 [1.00–1.00]	3.41 [3.40–3.43]
Cab fare per rider (Rs)	375 [373–377]	361 [359–364]	382 [380–385]	338 [336–340]	134 [132–136]

New Delhi (New Delhi Railway Station) · 2025-02-15 · N = 6,000 riders · sanity bound 0 min

Metric	Free-for-all	FCFS app	Market	Throughput	RAHA
Worst-off wait (min)	420 [420–420]	339 [337–341]	420 [420–420]	413 [412–414]	328 [325–330]
Mean wait (min)	226.0 [225.9–226.1]	226.0 [225.9–226.1]	226.0 [225.9–226.2]	226.0 [225.9–226.1]	226.0 [225.9–226.2]
Wheelchair p95 wait (min)	395 [392–397]	326 [324–329]	393 [391–396]	406 [404–407]	310 [307–314]
Turned away at doors	19,982 [19,971–19,993]	0 [0–0]	0 [0–0]	0 [0–0]	0 [0–0]
Peak door density (p/m²)	16.15 [16.14–16.16]	5.00 [5.00–5.00]	5.00 [5.00–5.00]	5.00 [5.00–5.00]	1.00 [1.00–1.00]
Unserved at horizon	2,500 [2,500–2,500]	2,500 [2,500–2,500]	2,500 [2,500–2,500]	2,500 [2,500–2,500]	2,500 [2,500–2,500]

Mumbai (Kurla station) · 2025-08-19 · N = 5,000 riders · sanity bound 133 min

Metric	Free-for-all	FCFS app	Market	Throughput	RAHA
Worst-off wait (min)	238 [238–238]	193 [192–194]	240 [240–240]	236 [235–236]	176 [175–176]
Mean wait (min)	73.6 [73.6–73.7]	80.3 [80.2–80.5]	79.5 [79.3–79.6]	80.2 [80.0–80.3]	61.6 [61.4–61.8]
Wheelchair p95 wait (min)	189 [186–191]	168 [166–170]	204 [202–206]	231 [230–232]	140 [137–143]
Deadlines met (share)	0.51 [0.51–0.51]	0.41 [0.41–0.41]	0.52 [0.52–0.52]	0.42 [0.42–0.42]	0.72 [0.72–0.73]
Turned away at doors	10,109 [10,087–10,127]	0 [0–0]	0 [0–0]	0 [0–0]	0 [0–0]
Peak door density (p/m²)	17.92 [17.82–18.04]	2.00 [2.00–2.00]	2.00 [2.00–2.00]	2.00 [2.00–2.00]	1.00 [1.00–1.00]
Unserved at horizon	900 [897–903]	1,242 [1,232–1,253]	1,196 [1,186–1,208]	1,235 [1,226–1,246]	131 [122–140]
Riders per cab	1.00 [1.00–1.00]	1.00 [1.00–1.00]	1.00 [1.00–1.00]	1.00 [1.00–1.00]	3.64 [3.62–3.66]
Cab fare per rider (Rs)	260 [259–262]	202 [201–202]	317 [315–319]	159 [159–159]	124 [123–126]

Singapore (Jurong East station) · 2024-09-25 · N = 9,000 riders · sanity bound 48 min

Metric	Free-for-all	FCFS app	Market	Throughput	RAHA
Worst-off wait (min)	198 [197–199]	86 [82–89]	115 [110–120]	84 [80–87]	66 [63–68]
Mean wait (min)	30.1 [30.0–30.2]	13.6 [13.5–13.6]	13.6 [13.6–13.6]	13.5 [13.5–13.5]	13.1 [13.1–13.1]
Wheelchair p95 wait (min)	117 [114–121]	65 [62–69]	68 [65–72]	64 [61–67]	46 [44–48]
Deadlines met (share)	0.68 [0.68–0.69]	0.92 [0.92–0.93]	0.92 [0.92–0.92]	0.93 [0.92–0.93]	0.97 [0.97–0.97]
Turned away at doors	24,292 [24,136–24,441]	0 [0–0]	0 [0–0]	0 [0–0]	0 [0–0]
Peak door density (p/m²)	25.25 [25.13–25.37]	2.40 [2.40–2.40]	2.40 [2.40–2.40]	2.40 [2.40–2.40]	1.00 [1.00–1.00]
Unserved at horizon	0 [0–0]	0 [0–0]	0 [0–0]	0 [0–0]	0 [0–0]
Riders per cab	1.00 [1.00–1.00]	1.00 [1.00–1.00]	1.00 [1.00–1.00]	1.00 [1.00–1.00]	3.90 [3.90–3.90]
Cab fare per rider (S$)	18.99 [18.96–19.01]	19.19 [19.16–19.23]	20.48 [20.43–20.53]	14.57 [14.55–14.59]	3.72 [3.71–3.73]

Madrid (Atocha station) · 2025-04-28 · N = 6,000 riders · sanity bound 42 min

Metric	Free-for-all	FCFS app	Market	Throughput	RAHA
Worst-off wait (min)	223 [222–224]	105 [103–107]	198 [198–199]	218 [216–220]	87 [86–88]
Mean wait (min)	45.3 [45.2–45.4]	43.0 [42.8–43.2]	41.9 [41.7–42.0]	43.5 [43.4–43.7]	32.4 [32.3–32.5]
Wheelchair p95 wait (min)	130 [127–134]	84 [82–86]	137 [133–142]	208 [206–209]	55 [53–57]
Deadlines met (share)	0.79 [0.79–0.79]	0.75 [0.75–0.75]	0.78 [0.78–0.78]	0.75 [0.75–0.75]	0.93 [0.93–0.93]
Turned away at doors	11,626 [11,582–11,675]	0 [0–0]	0 [0–0]	0 [0–0]	0 [0–0]
Peak door density (p/m²)	12.92 [12.82–13.01]	2.25 [2.25–2.25]	2.25 [2.25–2.25]	2.25 [2.25–2.25]	1.00 [1.00–1.00]
Unserved at horizon	0 [0–0]	0 [0–0]	0 [0–0]	0 [0–0]	0 [0–0]
Riders per cab	1.00 [1.00–1.00]	1.00 [1.00–1.00]	1.00 [1.00–1.00]	1.00 [1.00–1.00]	3.85 [3.84–3.86]
Cab fare per rider (EUR)	19.32 [19.20–19.43]	15.42 [15.19–15.66]	22.93 [22.79–23.09]	8.64 [8.63–8.65]	9.73 [9.70–9.76]

Constraint 1 test: internet cut at 12:33 in the Madrid blackout, worst-off wait · turned away

Phone reach after the cut	Free-for-all	FCFS app (cloud)	RAHA (cloud)	RAHA (edge node)
1 (no cut)	223 min · 11,620 turned away	106 min · 0 turned away	88 min · 0 turned away	88 min · 0 turned away
0.7	223 min · 11,627 turned away	225 min · 11,623 turned away	223 min · 11,625 turned away	83 min · 0 turned away
0.3	224 min · 11,637 turned away	223 min · 11,637 turned away	223 min · 11,634 turned away	81 min · 0 turned away
0	229 min · 12,900 turned away	229 min · 12,903 turned away	229 min · 12,900 turned away	99 min · 0 turned away

A coordinator that lives in the cloud collapses into free-for-all the moment the internet goes. RAHA's local-first edge node keeps every guarantee at every level of phone reach.

Constraint 3 test: provider feed changes shape mid-incident (Singapore line closure)

Feed mutation (from 10:30)	Naive parser: empty buses · turned away · worst wait	RAHA adapter: empty buses · turned away · worst wait
rename	275 · 0 · 219 min	3 · 0 · 65 min
nest	275 · 0 · 219 min	3 · 0 · 65 min
retype	275 · 0 · 219 min	3 · 0 · 65 min
optional	90 · 0 · 125 min	3 · 0 · 65 min
garbage	92 · 3,938 · 143 min	3 · 0 · 65 min
semantic	2 · 1,385 · 68 min	3 · 0 · 65 min

Rename, nest, retype, optional, garbage and semantic drift: the naive parser reads NaN, throws, or trusts a field whose meaning changed, so buses leave empty or are overbooked. RAHA's adapter (aliases, coercion, invariants, last-known-good values and a local registry) keeps every seat count correct.

Sensitivity: do the conclusions survive?

Stress test (Chennai)	Free-for-all worst wait	FCFS worst wait	RAHA worst wait	RAHA turned away
No door-crush delay for free-for-all	226	220	200	0
Demand ×0.5	202	186	108	0
Demand ×2	240	232	230	0
Fleet ×0.5	239	225	226	0

Live judge check. Pressing J in the simulator re-runs both constraint tests in the judge's own browser:

- blocks a real call to example.com
- reports the offline cache status
- runs all six API mutations against the actual parser code
- simulates the internet cut
- proves determinism (same seed, same SHA-256)

7. Viability and business model

RAHA is free for riders and paid for by the operators who carry the legal and reputational cost of a stranded station. The simulation is part of the product: operators use it to size bridging reserves and holding areas before an incident, and the same engine runs live during one.

Who pays	Model	Why
Rail, metro and bus operators (B2G)	per-station annual licence + incident drills	Railways is already funding holding areas and AI monitoring at 76 stations; RAHA is the capacity-linked release and bridging layer on top. Pender et al. found few agencies keep bus reserves: RAHA's drills size them.
Campuses and employers (B2B circles)	per-member monthly fee	Trusted circles (Zimride's model, which became Lyft) for students and shift workers: guaranteed rides home when the network fails, cheaper than solo cabs.
Delivery and bike-taxi platforms	per geofenced dispatch lead	Idle partners inside the geofence get paid trips (urgent riders, essentials drops) exactly when demand spikes.
Event organisers (Kumbh, stadiums, concerts)	per-event deployment	Capacity-linked boarding windows and holding-area countdowns for planned surges like the one at New Delhi.

Go to market. Pilot at Chennai (Tambaram station) with Southern Railway Chennai Division and MTC via SRMIST (riders, data and a live corridor next door), then the 76 holding-area stations. Riders adopt Koottu for free because sharing is cheaper than riding alone during surge.

Evidence the market pays. Transit software sold to governments is a real market: Via listed at about $3.5bn with over 90% of revenue from public agencies. Indian state transport buys digital services: Chalo works with TNSTC, DTC and KSRTC. Station crowd tools are tendered today: one division's information-board tender alone is about Rs 2.13 cr.

Risks. Long public-sector sales cycles; two-wheeler and multi-stop pooling rules vary by state; RAHA cannot switch surge off, only make it hurt less; fare quotes are entered by riders, never scraped.

8. Honest limits

- RAHA cannot switch surge pricing off. It makes surge hurt less.
- Rider populations are synthetic, drawn from real counts and distributions. Assumptions are labelled T4 and stress-tested.
- The 2× surge cap is a central guideline. No Tamil Nadu notification adopting it was found, so without a verified base fare the check says "cannot check".
- Grouping is a heuristic, not an optimum. Two-wheeler rides and multi-stop pooling depend on state permits.
- Station footfall, fleet and fare figures that were not published are explicit assumptions; each one is listed in the simulator's Data tab.

9. Key sources

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